Simple Statistics Predict Random-Walk Cover Times on Small Graphs

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Abstract

The expected cover time $C(G)$ of a simple random walk on a connected graph is classically bounded by $2m(n-1)$, which is loose for small graphs. We ask whether a regression on a few cheap statistics (vertex count n, edge count m, maximum degree Δ, diameter D) can predict $C(G)$ more accurately for connected graphs with $n \leq 30$, and where such a predictor fails. We generate 168 graphs from 11 families, estimate $C(G)$ by vectorised Monte Carlo (1000 walks per start vertex, worst start), validate the estimator against exact linear-system values for the 68 graphs with $n \leq 10$, and evaluate log-linear predictors with leave-one-family-out (LOFO) cross-validation. The classic bound overestimates $C(G)$ by a median factor of about 3.6 (median relative error 0.53). A three-feature log-linear model $\log C \approx a + b \log n + c \log m + d \log D$ reduces LOFO median relative error to 0.19 (95% bootstrap CI $[0.16, 0.24]$), and adding $\log \Delta$ to give four features reduces it to 0.16. Error is strongly family-dependent: the four-feature model is accurate on paths, cycles, random trees and sparse random graphs (median error 0.04–0.09), but the three-feature model is poor on cycles (0.58), and both models are poor on grids, lollipops, $G(n,p)$ and the complete graph (median error 0.5–0.7), where cover time is governed by bottlenecks or density effects that these global statistics do not capture. All conclusions are specific to this small-graph synthetic benchmark.

1 Introduction

For a random walk on a connected undirected graph, the cover time is the expected number of steps needed to visit every vertex, maximised over start vertices. Aleliunas et al. (1979) showed that it is at most $2m(n-1)$, and later work (e.g. Feige, 1995; Lovász, 1993) gives sharper asymptotic characterisations in terms of hitting times. These results are asymptotic or worst-case over graph classes, so for a specific graph with $n \leq 30$ they can be far from the true value. A practitioner who needs a quick estimate of cover time for a small network would benefit from a predictor that uses only a handful of easily computed statistics and whose error is known per graph type. We provide such an empirical predictor and measure its reliability.

Research question. For connected graphs with at most 30 vertices, can a few cheap statistics (n, m, mean/max degree, diameter) predict $C(G)$ more accurately than $2m(n-1)$, and how does the error depend on graph type?

Contributions. (i) A reproducible Monte Carlo benchmark of $C(G)$ over 168 graphs, validated against exact values for $n \leq 10$. (ii) A comparison of the classic bound with simple log-linear predictors under family-held-out evaluation. (iii) A family-level error analysis that identifies where the predictors fail.

2 Method

Estimand. $C(G) = \max_s \mathbb{E}_s[\tau_{\text{cover}}]$. For $n \leq 8$ all starts are simulated; for larger n we simulate four candidate starts (both ends of a double-sweep diameter path, a maximum-degree vertex, and one random vertex) and take the largest mean. Selecting the maximum over starts introduces a small upward bias; we keep it because it targets the definition.

Monte Carlo. $R = 1000$ independent walks per start are simulated in vectorised NumPy until every vertex is visited. The standard error of the mean is reported. The median relative SE is 0.017 (max 0.032).

Exact validation. For $n \leq 10$ we solve the linear system for the expected remaining cover time over states (v, S), $v \in S$, with sparse LU, and take the maximum over starts. Across 68 graphs, mean

$|C_{\mathrm{MC}}/C_{\mathrm{exact}} - 1| = 0.013$ (seed 0) and 0.015 (seed 1); the largest $|z|$ is 2.1 and 2.6, respectively, and 98.5% (seed 0) / 95.6% (seed 1) of estimates fall within two standard errors of the exact value.

Graphs. Eight deterministic families (path, cycle, star, complete, $\sqrt{n} \times \sqrt{n}$ grid, lollipop, barbell, binary tree) and three random families (random recursive tree, $G(n,p)$ with $p = 2\ln n/n + 0.05$ conditioned on connectivity, and random tree plus $\lfloor n/4 \rfloor$ extra edges). Sizes $n \in \{4, 6, 8, 10, 12, 16, 20, 24, 30\}$; random families get 2 or 4 instances per size.

Predictors. All are OLS fits of $\log \hat{C}$ on:

- `bound_scale`: $\log(2m(n-1))$ with one fitted scale and intercept;
- `n_m`: $\log n$, $\log m$;
- `n_m_D`: $\log n$, $\log m$, $\log D$;
- `full`: additionally $\log \Delta$.

Evaluation uses leave-one-family-out CV (fit on 10 families, predict the held-out one), so each prediction is made for a family not seen in training. Uncertainty over graphs is a 2000-sample bootstrap of the median LOFO error. Relative error is $|\hat{C}_{\mathrm{pred}} - \hat{C}_{\mathrm{MC}}|/\hat{C}_{\mathrm{MC}}$.

3 Experiments

Table 1 gives the main results (seed 0; seed 1 in brackets). Table 2 gives per-family LOFO median errors for seed 0.

Table 1: Predictor accuracy over 168 graphs. LOFO = leave-one-family-out. CI = 95% bootstrap interval for the LOFO median (seed 0).

Predictor	In-sample median err.	LOFO median err.	LOFO 90th pct.
Classic bound $2m(n-1)$ (scaled)	0.48 [0.50]	0.53 [0.56] (CI 0.50–0.61)	2.81 [2.83]
$\log n, \log m$	0.20 [0.20]	0.23 [0.26] (CI 0.19–0.32)	0.83 [0.84]
$\log n, \log m, \log D$	0.14 [0.14]	0.19 [0.18] (CI 0.16–0.24)	0.76 [0.73]
$+ \log \Delta$	0.13	0.16 (CI 0.11–0.20)	0.74

Table 2: Per-family LOFO median relative error (seed 0), for the three-feature model (`n_m_D`), the four-feature model (`full`), and the classic bound (scaled).

Family	n_m_D	full	bound (scaled)
sparse random	0.09	0.09	0.38
random tree	0.13	0.07	0.52
path	0.13	0.04	0.62
binary tree	0.13	0.25	0.47
complete	0.20	0.48	5.26
barbell	0.27	0.31	0.59
star	0.52	0.16	0.12
$G(n,p)$	0.54	0.66	1.90
cycle	0.58	0.06	0.41
lollipop	0.62	0.53	0.68
grid	0.71	0.67	0.38

Fitted exponents for `n_m_D` on all data are $b_n = 1.35$, $b_m = 0.07$, $b_D = 0.78$ (seed 1: 1.33, 0.08, 0.77). The edge exponent is near zero once n and D are known, so $C(G)$ is governed mainly by size and diameter in this benchmark. The classic bound has the right order of growth in m and n but overestimates by a

graph-dependent factor: the ratio $C/[2m(n-1)]$ has median 0.27 and ranges from 0.005 (complete) to 0.64 (path). Residuals of `n_m_D` correlate weakly with Δ/n ($r = 0.30$), and not with $\log(m/n)$ ($r = -0.05$) or D/n ($r = 0.03$), which is consistent with the four-feature model's gain coming from degree heterogeneity.

4 Discussion and limitations

A three-feature log-linear model roughly halves the error of the classic bound on held-out families, and this holds on both seeds. The four-feature model is reliable on paths, cycles, random trees and sparse random graphs, where the cover time is driven mainly by size and diameter (the three-feature model is not reliable on cycles). It is unreliable on grids, lollipops and barbells, which contain bottlenecks that the global statistics do not see, and on dense graphs ($G(n,p)$, complete), where the within-family relationship differs from the sparse families and LOFO extrapolation fails. The 90th-percentile errors (about 0.75) show that individual predictions can be off by a large factor, so the predictor should be used only as a rough estimate with an error band.

Limitations: (1) the benchmark is synthetic and small ($n \leq 30$), and the four-feature model's coefficients are fitted to these families only; (2) the target is the maximum over starts, which for $n > 8$ is taken over four candidate starts, so it may slightly overstate the true maximum; (3) only two Monte Carlo seeds of graph generation were run, with one simulation seed per graph; (4) the families were chosen by us, and LOFO errors depend on this choice; (5) we did not compare to other baseline predictors (e.g. hitting-time-based Matthews bounds or spectral quantities), so claims of superiority are relative only to the classic bound and to log-linear fits on these statistics. Exact validation was limited to $n \leq 10$.

5 Conclusion

On small connected graphs, a log-linear predictor in n, m and diameter (optionally Δ) gives substantially better cover-time estimates than $2m(n-1)$, with median held-out relative error of about 0.16–0.19. Its error is strongly family-dependent: it is accurate on paths, cycles, random trees and sparse random graphs under the four-feature model, and not reliable on bottleneck or dense graphs.

Reproducibility

Code: `cover_experiment.py` (simulation and exact solver), `analyze.py` (fits and CV). Commands: `python3 cover_experiment.py --seed 0 --R 1000 --out results.csv`; `python3 analyze.py results.csv > analysis.txt`; the same with `--seed 1` for the robustness run. Seeds: graph generation uses `seed`; Monte Carlo uses `seed+1000`; bootstrap uses 12345. Software: Python 3.11, NumPy 2.4, SciPy 1.17, pandas. Runtime: about 1–2 minutes per seed on CPU.

References

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